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Analysis on an HDG Method for the *p*-Laplacian Equations

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Abstract
In Cockburn and Shen (SIAM J Sci Comput 38(1):A545–A566, 2016) the authors propose the first hybridizable discontinuous Galerkin method (HDG) for the p-Laplacian equation. Several iterative algorithms are developed and tested. The main purpose of this paper is to provide rigorous error estimates for the proposed HDG method. We first develop the error estimates based on general polyhedral/polygonal triangulations, under standard regularity assumption of the solution, the convergence analysis is presented for $1 < p < 2$ and $p > 2$. Nevertheless, when p approaches to the limits ($p \to 1^+$ or $p \to \infty$), the convergence rate shows degeneration for both cases. Finally, this degeneration can be recovered if we use simplicial triangulation of the domain with sufficient large stabilization parameter for the method.

Keywords Discontinuous Galerkin · p-Laplacian · Hybridization

Mathematics Subject Classification 65N30 · 65L12

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1 Introduction

In [15], the authors propose a new hybridizable discontinuous Galerkin method (HDG) for the p-Laplacian equation which can be written in the mixed form as:

$$\boldsymbol{q} + \nabla u = 0 \qquad \text{in } \Omega, \tag{1.1a}$$

$$\boldsymbol{\sigma} - |\boldsymbol{q}|^{p-2}\boldsymbol{q} = 0 \qquad \text{in } \Omega, \tag{1.1b}$$

$$\nabla \cdot \boldsymbol{\sigma} = f \qquad \text{in } \Omega \tag{1.1c}$$

$$u = g \qquad \text{on } \partial\Omega. \tag{1.1d}$$

Here $\Omega \in \mathrm{R}^2$ is a two-dimensional polyhedral domain. The unknown variables are u, $\boldsymbol{q}$ and $\boldsymbol{\sigma}$. Here we assume $f \in L^2(\Omega)$ and $g \in W^{1,1/p*}(\partial\Omega)$ where $1/p + 1/p* = 1$, to ensure the well-posedness of problem, see [2,26]. In addition, we have the stability estimate:

$$\|u\|_{W^{1,p}(\Omega)} \leq C\left[\|f\|_{L^2(\Omega)}^{\frac{1}{p-1}} + \|g\|_{W^{1-\frac{1}{p},p}(\partial\Omega)}\right]. \tag{1.2}$$

The nonlinear diffusion model appears in various applications in fluid [9,29,32,42] and solid mechanics [31,38]. We refer [15] for more details. In contrast with the linear case with $p = 2$, designing reliable and efficient numerical solvers for (1.1) remain an active and challenging research area. Glowinski and Marrocco [26] introduced the first finite element method for the $p-$Laplacian. a priori error estimates was developed to show the convergence of the numerical solution. The analysis was then improved by Chow [8]. It shows that with linear Lagrange elements, the numerical solution converges to the exact solution in the $W^{1,p}$ norm with the convergence order of $\frac{p}{2}$ $(1 < p < 2)$, $\frac{2}{p}$ $(p > 2)$ respectively. Later in [2,22] the authors successfully obtain optimal order of convergence. In [24,25] Farhloul and collaborators proposed and analyzed a mixed method using the lowest order Raviart–Thomas element. Their analysis provides convergence rate for both u and $\boldsymbol{\sigma}$. On the other hand, Berman and Ern first studied an IP-DG method for the nonlinear Laplacian problems in [4]. Soon in [3], the authors developed several analytical tools for the IP-DG methods for general nonlinear minimization problems and variational convergence of the numerical solution was proved. Later in Diening et al. [20] studied a local DG (LDG) method for the generalized $p-$Laplacian problem. a priori error estimates is established for the DG methods with specific order of convergence for any $p \in (1, \infty)$. In certain cases, it was shown that the LDG solution shares the optimal convergence rate with the CG method.

The HDG methods were first introduced in [10] for second order elliptic problems. Therein, it was shown that HDG methods can be implemented as efficiently as hybridized mixed methods. Later in [12] it was proven that the HDG methods do share with mixed methods their superior convergence properties while retaining the advantages typical of the DG methods such as flexible meshing, *hp*-adaptivity, local conservation. Due to these features, during last decade, there have been a blast of studies on HDG methods on a variety of linear and nonlinear problems such as Stokes flow [1,11,14,17,17]; Convection–diffusion [5,23,37]; wave propagations [13,35]; Maxwell equations [7]; Helmholtz equation [19,27,28,34]; linear and nonlinear elasticity [16,39,43]; Incompressible Navier–Stokes equation and magnetohydrodynamics [40,41]; third order KdV type equations [6,21]; nonlinear conservation laws [33,36]. Amazingly, with appropriate choices of the finite element spaces and the stabilization strategy, HDG provides optimal order of convergence for all the problems mentioned above.

Recently in [15], for the first time, Cockburn and Shen considered a hybridizable DG method for the p-Laplacian problem with high order polynomial spaces. They numerically

tested the proposed HDG method which shows that the HDG method can achieve high order accuracy and even superconvergent solutions when the exact solution is sufficiently smooth. They also tested the method on less regular examples and the numerical solution remains convergent to the exact solution. In contrast, the goal of this paper is to provide the first a priori error estimates for the HDG method proposed in [15]. Namely, we first present a suboptimal convergence analysis for the method on general polygonal triangulations. In addition, we show that with simplicial triangulation, the HDG solution can be sufficiently close to the CG solution if the stabilization parameter is chosen large enough. Consequently, the HDG solution shares the same optimal convergence property as the CG solution.

This paper is organized as follows. In Sect. 2 we present the method introduced in [15] and some auxiliary estimates for the analysis. In Sect. 3 we present the main error estimates for the proposed method on general polygonal meshes. In Sect. 4, we establish a better convergence result based on simplicial meshes.

2 Preliminary Results

In this section, we present the HDG method for the model problem (1.1) together with some auxiliary results for the analysis in later sections. To define the HDG method, we consider conforming quasi-uniform triangulation $\mathcal{T}_h$ of Ω made of shape-regular *polygon*. We denote by $\mathcal{E}_h^0$ the set of all interior faces F of all elements $K \in \mathcal{T}_h$, $\mathcal{E}_h^\partial$ the set of the boundary faces F on $\partial\Omega$ and we denote $\mathcal{E}_h = \mathcal{E}_h^0 \cup \mathcal{E}_h^\partial$. We also set $\partial\mathcal{T}_h := \{\partial K : K \in \mathcal{T}_h\}$. For scalar-valued functions ϕ and ψ, we write

$$(\phi, \psi)_{\mathcal{T}_h} := \sum_{K\in\mathcal{T}_h} (\phi, \psi)_K, \quad \langle \phi, \psi\rangle_{\partial\mathcal{T}_h} := \sum_{K\in\mathcal{T}_h} \langle \phi, \psi\rangle_{\partial K}.$$

Here $(\cdot,\cdot)_D$ denotes the integral over the domain $D \subset \mathrm{R}^2$, and $\langle\cdot,\cdot\rangle_D$ denotes the integral over $D \subset \mathrm{R}^1$. For vector-valued functions we define $(\boldsymbol{\phi}, \boldsymbol{\psi})_{\mathcal{T}_h} := \sum_{i=1}^2 (\phi_i, \psi_i)_{\mathcal{T}_h}$. For each $K \in \mathcal{T}_h$, h_K denotes the diameter of K and we set $h = \max_{K\in\mathcal{T}_h} h_K$. Similarly, h_F denotes the length of a face $F \in \mathcal{E}_h$. The HDG method in [15] seeks an approximation $(\boldsymbol{\sigma}_h, \boldsymbol{q}_h, u_h, \widehat{u}_h)$ to the exact solution $(\boldsymbol{\sigma}, \boldsymbol{q}, u, u|_{\mathcal{E}_h})$ in the finite dimensional space $\boldsymbol{V}_h \times \boldsymbol{V}_h \times W_h \times M_h(g)$ satisfying the system:

$$(\boldsymbol{q}_h\,,\ \boldsymbol{v}_1)_{\mathcal{T}_h} - (u_h\,,\ \nabla\cdot\boldsymbol{v}_1)_{\mathcal{T}_h} + \langle\widehat{u}_h\,,\ \boldsymbol{v}_1\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = 0, \tag{2.1a}$$

$$(\boldsymbol{\sigma}_h\,,\ \boldsymbol{v}_2)_{\mathcal{T}_h} - (|\boldsymbol{q}_h|^{p-2}\boldsymbol{q}_h\,,\ \boldsymbol{v}_2)_{\mathcal{T}_h} = 0, \tag{2.1b}$$

$$-(\boldsymbol{\sigma}_h\,,\ \nabla w)_{\mathcal{T}_h} + \langle\widehat{\boldsymbol{\sigma}}_h\cdot\boldsymbol{n}\,,\ w\rangle_{\partial\mathcal{T}_h} = (w\,,\ f)_{\mathcal{T}_h}, \tag{2.1c}$$

$$\langle\widehat{\boldsymbol{\sigma}}_h\cdot\boldsymbol{n},\quad \mu\rangle_{\partial\mathcal{T}_h} = 0, \tag{2.1d}$$

for all $(\boldsymbol{v}_1, \boldsymbol{v}_2, w.\mu) \in \boldsymbol{V}_h \times \boldsymbol{V}_h \times W_h \times M_h(0)$, where the finite element spaces are defined as:

$$\begin{aligned}
\boldsymbol{V}_h &:= \{\boldsymbol{v} \in [L^2(\Omega)]^2 : \boldsymbol{v}|_K \in [P^k(K)]^2 \quad \forall K \in \mathcal{T}_h\},\\
W_h &:= \{w \in L^2(\Omega) : w|_K \in P^k(K) \quad \forall K \in \mathcal{T}_h\},\\
M_h &:= \{\mu \in L^2(\mathcal{E}_h) : \mu|_F \in P^k(F) \quad \forall F \in \mathcal{E}_h\},\\
M_h(g) &:= \{\lambda \in M_h : \lambda|_{\partial\Omega} = g_h\},
\end{aligned}$$

here g_h denotes the conforming interpolant of g on $\partial\Omega$. Here $P^k(D)$ denotes the space of polynomials of degree no more than $k (k \geq 1)$ on D. Finally, the system is complete with the

following definition of the *numerical flux*:

$$\widehat{\boldsymbol{\sigma}}_h \cdot \boldsymbol{n} = \boldsymbol{\sigma}_h \cdot \boldsymbol{n} + \tau(u_h - \widehat{u}_h), \quad \text{on } \partial\mathcal{T}_h. \tag{2.2}$$

Here the stabilization parameter τ is piecewise positive constant on each edge $F \in \partial\mathcal{T}_h$. In [15], the authors show that the above system is well-posed if τ is strictly positive on each edge.

On each $K \in \mathcal{T}_h$, with any $(w, \mu) \in L^2(K) \times L^2(\partial K)$, we define the local projection $\boldsymbol{Q}_h(w, \mu) \in [P^k(K)]^2$ satisfying:

$$(\boldsymbol{Q}_h(w, \mu),\ \boldsymbol{v})_K - (w, \nabla \cdot \boldsymbol{v})_K + \langle \mu,\ \boldsymbol{v} \cdot \boldsymbol{n} \rangle_{\partial K} = 0, \quad \forall \boldsymbol{v} \in [P^k(K)]^2. \tag{2.3}$$

It is obvious that the above projection is well-defined. In addition, (2.1a) can be written as:

$$\boldsymbol{q}_h = \boldsymbol{Q}_h(u_h, \widehat{u}_h) \quad \text{on } \mathcal{T}_h.$$

Define $\mathcal{J}_h : W_h \times M_h \to \mathrm{R}$ by

$$\mathcal{J}_h(w, \mu) = \frac{1}{p}(|\boldsymbol{Q}_h(w, \mu)|^p, 1)_{\mathcal{T}_h} + \frac{1}{2}\langle \tau(w - \mu)\,,\ w - \mu \rangle_{\partial\mathcal{T}_h} - (f\ ,\ w)_{\mathcal{T}_h}. \tag{2.4}$$

The system is equivalent as the following minimization problem [15]:

$$(u_h, \widehat{u}_h) = \underset{(w,\mu)\in W_h \times M_h(g)}{\arg\min} \mathcal{J}_h(w, \mu). \tag{2.5}$$

In [15], it is shown that the minimizer is unique if the stabilization function τ is strictly positive on $\partial\mathcal{T}_h$. In addition, it is shown that the above functional is strictly convex over $W_h \times M_h(g)$. Furthermore, it is Gateaux differentiable with

$$\begin{aligned} \mathcal{J}_h'(w, \mu)(y, \nu) := &(|\boldsymbol{Q}_h(w, \mu)|^{p-2}\boldsymbol{Q}_h(w, \mu)\,,\ \boldsymbol{Q}_h(y, \nu))_{\mathcal{T}_h} \\ &+ \langle \tau(w - \mu)\,,\ y - \nu \rangle_{\partial\mathcal{T}_h} - (f\ ,\ y)_{\mathcal{T}_h}, \end{aligned} \tag{2.6}$$

for all $(w, \mu) \in W_h \times M_h(g)$, $(y, \nu) \in W_h \times M_h(0)$.

Next we introduce the projections and their approximation properties which are needed in the analysis. On each $K \in \mathcal{T}_h$, let Π_W, Π_V denote the standard L^2-projections onto $P^k(K)$, $[P^k(K)]^2$ respectively. Similarly, for each $F \in \mathcal{E}_h$, Π_M denotes the standard L^2-projection onto $P^k(F)$. With these projections together with (2.3), we can easily obtain that

$$\boldsymbol{Q}_h(\Pi_w u, \Pi_M u) = \Pi_V \boldsymbol{q}. \tag{2.7}$$

In the following analysis, we first gather some analitical tools for our main result. The first one is in [2, Lemma 2.1]:

Lemma 2.1 *For all $p > 1$ and $\delta > 0$, there exist positive constants C_1 and C_2 such that for all $\boldsymbol{\psi}, \boldsymbol{\eta} \in \mathrm{R}^2$, $\boldsymbol{\psi} \neq \boldsymbol{\eta}$,*

$$||\boldsymbol{\psi}|^{p-2}\boldsymbol{\psi} - |\boldsymbol{\eta}|^{p-2}\boldsymbol{\eta}| \leq C_1|\boldsymbol{\psi} - \boldsymbol{\eta}|^{1-\delta}(|\boldsymbol{\psi}| + |\boldsymbol{\eta}|)^{p-2+\delta}, \tag{2.8a}$$

$$(|\boldsymbol{\psi}|^{p-2}\boldsymbol{\psi} - |\boldsymbol{\eta}|^{p-2}\boldsymbol{\eta}) \cdot (\boldsymbol{\psi} - \boldsymbol{\eta}) \geq C_2|\boldsymbol{\psi} - \boldsymbol{\eta}|^{2+\delta}(|\boldsymbol{\psi}| + |\boldsymbol{\eta}|)^{p-2-\delta}. \tag{2.8b}$$

The second lemma is analogous to [2, Lemma 2.2]. For functions $\boldsymbol{\eta}, \boldsymbol{\psi} \in [L^p(\Omega)]^2$ and $\sigma > 1$ we first define

$$|\boldsymbol{\eta}|_{(\boldsymbol{\psi},p,\sigma)} := \int_\Omega (|\boldsymbol{\psi}| + |\boldsymbol{\eta}|)^{p-\sigma}|\boldsymbol{\eta}|^\sigma dx.$$

Lemma 2.2 *Let $\boldsymbol{\psi}, \boldsymbol{\eta} \in [L^p(\Omega)]^2$ and $\sigma > 1$, for $p \in (1, \sigma]$ we have*

$$|\boldsymbol{\eta}|^{\frac{\sigma}{p}}_{(\boldsymbol{\psi},p,\sigma)} \leq \|\boldsymbol{\eta}\|^{\sigma}_{L^p(\Omega)} \leq C\big(\|\boldsymbol{\psi}\|_{L^p(\Omega)} + \|\boldsymbol{\eta}\|_{L^p(\Omega)}\big)^{\sigma-p}|\boldsymbol{\eta}|_{(\boldsymbol{\psi},p,\sigma)}, \tag{2.9a}$$

and for $p \in [\sigma, \infty)$,

$$\|\boldsymbol{\eta}\|^{p}_{L^p(\Omega)} \leq |\boldsymbol{\eta}|_{(\boldsymbol{\psi},p,\sigma)} \leq C\big(\|\boldsymbol{\psi}\|_{L^p(\Omega)} + \|\boldsymbol{\eta}\|_{L^p(\Omega)}\big)^{p-\sigma}\|\boldsymbol{\eta}\|^{\sigma}_{L^p(\Omega)}, \tag{2.9b}$$

This result is a general version of the Lemma 2.2 in [2]. The proof is almost identical as the original version.

Lemma 2.3 *Let $u \in L^p(K)$ with $p \geq 1$, $K \in \mathcal{T}_h$, $\Pi_W u$ is the standard L^2-projection of u on $P^k(K)$, then we have*

$$\|\Pi_W u\|_{L^p(K)} \leq C\|u\|_{L^p(K)},$$

here C is independent of the diameter h_K of K.

Proof Let $\{\phi_1, \phi_2, \ldots, \phi_N\}$ be an orthonormal basis for $P^k(K)$ with respect to the standard L^2 inner product over K. For $u \in L^p(K)$, we can write

$$\Pi_W u = \sum_{i=1}^{N} c_i\phi_i, \quad \text{where} \quad c_i = (u, \phi_i)_K.$$

Therefore, we have

$$\begin{aligned}\|\Pi_W u\|_{L^p(K)} &= \|\sum_{i=1}^{N} c_i\phi_i\|_{L^p(K)} \\ &\leq \sum_{i=1}^{N}|c_i|\|\phi_i\|_{L^p(K)} \leq Ch^{\frac{2}{p}-1}\sum_{i=1}^{N}|c_i|\|\phi_i\|_{L^2(K)} = C_1h^{\frac{2}{p}-1}\sum_{i=1}^{N}|c_i|.\end{aligned}$$

Here we used the discrete inverse inequality for the polynomial space $P^k(K)$. Similarly, by the definition of c_i, we have

$$|c_i| = (u, \phi_i)_K \leq \|u\|_{L^p(K)}\|\phi_i\|_{L^q(K)} \leq C_2h_K^{\frac{2}{q}-1}\|u\|_{L^p(K)}.$$

Here q satisfies $\frac{1}{p} + \frac{1}{q} = 1$. Combining the above two estimates we have

$$\|\Pi_W u\|_{L^p(K)} \leq C_1C_2Nh_K^{\frac{2}{p}+\frac{2}{q}-2}\|u\|_{L^p(K)} \leq \mathcal{C}\|u\|_{L^p(K)}.$$

Here $\mathcal{C}$ depends on the shape of K and the dimension of the space $P^k(K)$ but is independent of h_K. This completes the proof. □

By the above estimate together with the stability estimate (1.2) we have:

$$\|\Pi_V \boldsymbol{q}\|_{L^p(\Omega)} \leq C\left[\|f\|^{\frac{1}{p-1}}_{L^2(\Omega)} + \|g\|_{W^{1-\frac{1}{p},p}(\partial\Omega)}\right]. \tag{2.10}$$

3 Error Estimates for General Polygonal Meshes

In this section, we present a priori error estimates of the method (2.1) on general polygonal meshes. In what follows, we only consider the case of $k = 1$ for the sake of simplicity. The analysis can be naturally extended to arbitrary polynomial degree. High order convergence rate can be obtained with stronger regularity assumptions which we will omit in this paper.

We define the projection of the errors as

$$e_u = \Pi_W u - u_h, \quad e_{\widehat{u}} = \Pi_M u - \widehat{u}_h, \quad \boldsymbol{e}_q = \Pi_V \boldsymbol{q} - \boldsymbol{q}_h.$$

In addition to the standard Sobolev norms, we define:

$$\|w\|^p_{L^p(\mathcal{T}_h)} := \sum_{K\in\mathcal{T}_h} \|w\|^p_{L^p(K)}, \quad \|w\|^2_{\partial\mathcal{T}_h} := \sum_{K\in\mathcal{T}_h} \|w\|^2_{\partial K},$$

for $p \in (1, \infty)$. We first present the following energy estimate:

Lemma 3.1 *With any $\delta \geq 0$, we have*

$$\begin{aligned} C\left(|\boldsymbol{e}_q|_{(\Pi_V\boldsymbol{q},p,2+\delta)} + \tau\|e_u - e_{\widehat{u}}\|^2_{\partial\mathcal{T}_h}\right) \\ \leq -\langle e_u - e_{\widehat{u}}\,,\, (\boldsymbol{\sigma} - \Pi_V\boldsymbol{\sigma})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V\boldsymbol{q}|^{p-2}\Pi_V\boldsymbol{q}\,,\, \boldsymbol{e}_q)_{\mathcal{T}_h} \\ + \langle\tau(\Pi_W u - \Pi_M u)\,,\, e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h}. \end{aligned}$$

Proof By (2.5), $(u_h, \widehat{u}_h)$ is the unique minimizer of $\mathcal{J}_h(\cdot,\cdot)$ on $W_h \times M_h(g)$. Hence we have

$$\mathcal{J}_h(u_h, \widehat{u}_h) - \mathcal{J}_h(\Pi_W u, \Pi_M u) \leq 0.$$

In addition, we can write the left hand side of the above inequality as follows:

$$\begin{aligned} &\mathcal{J}_h(u_h, \widehat{u}_h) - \mathcal{J}_h(\Pi_W u, \Pi_M u) \\ &\quad= \int_0^1 \mathcal{J}_h'(\Pi_W u + s(u_h - \Pi_W u), \Pi_M u + s(\widehat{u}_h - \Pi_M u))(-e_u, -e_{\widehat{u}})ds \\ &\quad= \int_0^1 \mathcal{J}_h'(\Pi_W u - se_u, \Pi_M u - se_{\widehat{u}})(-e_u, -e_{\widehat{u}})ds \\ &\quad= \int_0^1 \left(\mathcal{J}_h'(\Pi_W u, \Pi_M u) - \mathcal{J}_h'(\Pi_W u - se_u, \Pi_M u - se_{\widehat{u}})\right)(e_u, e_{\widehat{u}})ds \\ &\qquad - \mathcal{J}_h'(\Pi_W u, \Pi_M u)(e_u, e_{\widehat{u}}) \leq 0. \end{aligned}$$

Hence, we have the estimate:

$$\int_0^1 \left(\mathcal{J}_h'(\Pi_W u, \Pi_M u) - \mathcal{J}_h'(\Pi_W u - se_u, \Pi_M u - se_{\widehat{u}})\right)(e_u, e_{\widehat{u}})ds \leq \mathcal{J}_h'(\Pi_W u, \Pi_M u)(e_u, e_{\widehat{u}}).$$

The integral on the left hand side can be further expanded as:

$$\begin{aligned} &\int_0^1 \left(\mathcal{J}_h'(\Pi_W u, \Pi_M u) - \mathcal{J}_h'(\Pi_W u - se_u, \Pi_M u - se_{\widehat{u}})\right)(e_u, e_{\widehat{u}})ds \\ &= \int_0^1 (|\Pi_V\boldsymbol{q}|^{p-2}\Pi_V\boldsymbol{q} - |\Pi_V\boldsymbol{q} - s\boldsymbol{e}_q|^{p-2}(\Pi_V\boldsymbol{q} - s\boldsymbol{e}_q)\,,\, \boldsymbol{e}_q)_{\mathcal{T}_h} + s\langle\tau(e_u - e_{\widehat{u}})\,,\, e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h} ds \\ &\geq C\int_0^1\int_\Omega |s\boldsymbol{e}_q|^{2+\delta}(|\Pi_V\boldsymbol{q}| + |\Pi_V\boldsymbol{q} - s\boldsymbol{e}_q|)^{p-2-\delta}dxds + \frac{1}{2}\tau\|e_u - e_{\widehat{u}}\|^2_{\partial\Omega_h}. \end{aligned}$$

The last step is due to (2.8b). Further, notice that for $0 \le s \le 1$ we have

$$\frac{s}{2}(|\Pi_V \boldsymbol{q}| + |\boldsymbol{e}_q|) \le |\Pi_V \boldsymbol{q}| + |\Pi_V \boldsymbol{q} - s\boldsymbol{e}_q| \le 2(|\Pi_V \boldsymbol{q}| + |\boldsymbol{e}_q|).$$

This implies that

$$\begin{aligned}\int_0^1 \big(\mathcal{J}_h'(\Pi_W u, \Pi_M u) - \mathcal{J}_h'(\Pi_W u - s e_u, \Pi_M u - s e_{\widehat{u}})\big)(e_u, e_{\widehat{u}})ds \\ \ge C|\boldsymbol{e}_q|_{(\Pi_V \boldsymbol{q}, p, 2+\delta)} + \frac{\tau}{2}\|e_u - e_{\widehat{u}}\|^2_{\partial\Omega_h}.\end{aligned} \tag{3.1}$$

On the other hand, for $\mathcal{J}_h'(\Pi_W u, \Pi_M u)(e_u, e_{\widehat{u}})$ we start with its explicit expression:

$$\begin{aligned}\mathcal{J}_h'(\Pi_W u, \Pi_M u)(e_u, e_{\widehat{u}}) &= (|\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\ ,\ \boldsymbol{e}_q)_{\mathcal{T}_h} \\ &\quad + \langle \tau(\Pi_W u - \Pi_M u)\ ,\ e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h} - (f\ ,\ e_u)_{\mathcal{T}_h} \\ &= (\boldsymbol{\sigma}\ ,\ \boldsymbol{e}_q)_{\mathcal{T}_h} - (\nabla\cdot\boldsymbol{\sigma}\ ,\ e_u)_{\mathcal{T}_h} - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\ ,\ \boldsymbol{e}_q)_{\mathcal{T}_h} \\ &\quad + \langle \tau(\Pi_W u - \Pi_M u)\ ,\ e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h} \\ &= (\Pi_V \boldsymbol{\sigma}\ ,\ \boldsymbol{e}_q)_{\mathcal{T}_h} - (\nabla\cdot\Pi_V \boldsymbol{\sigma}\ ,\ e_u)_{\mathcal{T}_h} - (\nabla\cdot(\boldsymbol{\sigma} - \Pi_V \boldsymbol{\sigma})\ ,\ e_u)_{\mathcal{T}_h} \\ &\quad - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\ ,\ \boldsymbol{e}_q)_{\mathcal{T}_h} \\ &\quad + \langle \tau(\Pi_W u - \Pi_M u)\ ,\ e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h}.\end{aligned}$$

Notice that the system (2.1) is consistent so that the exact solution $(\boldsymbol{\sigma}, \boldsymbol{q}, u, u|_{\mathcal{E}_h})$ also satisfies the system (2.1). In particular, by (2.1a) with $\boldsymbol{v}_1 = \Pi_V \boldsymbol{\sigma}$ together with the orthogonal property of the projections, we arrive at the error equation as:

$$(\boldsymbol{e}_q\ ,\ \Pi_V \boldsymbol{\sigma})_{\mathcal{T}_h} - (e_u\ ,\ \nabla\cdot\Pi_V \boldsymbol{\sigma})_{\mathcal{T}_h} + \langle e_{\widehat{u}}\ ,\ \Pi_V \boldsymbol{\sigma}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} = 0.$$

Inserting above identity into the expression of $\mathcal{J}_h'(\Pi_W u, \Pi_M u)(e_u, e_{\widehat{u}})$ we have

$$\begin{aligned}\mathcal{J}_h'(\Pi_W u, \Pi_M u)(e_u, e_{\widehat{u}}) &= -\langle e_{\widehat{u}}\ ,\ \Pi_V \boldsymbol{\sigma}\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (\nabla\cdot(\boldsymbol{\sigma} - \Pi_V \boldsymbol{\sigma})\ ,\ e_u)_{\mathcal{T}_h} \\ &\quad - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\ ,\ \boldsymbol{e}_q)_{\mathcal{T}_h} \\ &\quad + \langle \tau(\Pi_W u - \Pi_M u)\ ,\ e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h} \\ &= -\langle e_u - e_{\widehat{u}}\ ,\ (\boldsymbol{\sigma} - \Pi_V \boldsymbol{\sigma})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} \\ &\quad - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\ ,\ \boldsymbol{e}_q)_{\mathcal{T}_h} \\ &\quad + \langle \tau(\Pi_W u - \Pi_M u)\ ,\ e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h}.\end{aligned}$$

The last step here is due to the integrating by parts, orthogonal property of the projection Π_V and the fact that $\boldsymbol{\sigma} \in \boldsymbol{H}(\mathrm{div}, \Omega)$ and $e_{\widehat{u}}$ is single-valued over $\mathcal{E}_h$ and $e_{\widehat{u}} = 0$ on $\partial\Omega$. The proof is complete with above identity and (3.1). □

Now we are ready to give our first main result:

Theorem 3.2 *If the exact solution $u \in H^2(\Omega) \cap W^{2,p}(\Omega)$, $\boldsymbol{q} \in W^{s,p}(\Omega)$, $\boldsymbol{\sigma} \in H^1(\Omega)$ with $s \ge 1$ and $t = \min\{s, 2\}$, then we have*

$$\text{for } p \in (1, 2] \quad \|\boldsymbol{e}_q\|_{L^p(\Omega)} \le Ch(\|\boldsymbol{\sigma}\|_{H^1(\Omega)} + \|u\|_{H^2(\Omega)}) + Ch^{t(p-1)}\|\boldsymbol{q}\|^{p-1}_{W^{t,p}(\Omega)}, \tag{3.2a}$$

$$\text{for } p \in (2, \infty) \quad \|\boldsymbol{e}_q\|_{L^p(\Omega)} \le Ch^{\frac{2}{p}}(\|\boldsymbol{\sigma}\|_{H^1(\Omega)} + \|u\|_{H^2(\Omega)}) + Ch^{\frac{t}{p-1}}\|\boldsymbol{q}\|_{W^{t,p}(\Omega)}. \tag{3.2b}$$

Proof We first consider the case $p \in (1, 2]$. In this case, we take $\sigma = 2$ in (2.9a) with $(\boldsymbol{\psi}, \boldsymbol{\eta}) = (\Pi_V \boldsymbol{q}, \boldsymbol{e}_q)$. This implies that

$$\|\boldsymbol{e}_q\|^2_{L^p(\Omega)} \le C(\|\Pi_V \boldsymbol{q}\|_{L^p(\Omega)} + \|\boldsymbol{e}_q\|_{L^p(\Omega)})^{2-p} |\boldsymbol{e}_q|_{(\Pi_V \boldsymbol{q}, p, 2)} \le C|\boldsymbol{e}_q|_{(\Pi_V \boldsymbol{q}, p, 2)}.$$

Here C depends on the source terms f, g due to (1.2) and Lemma 2.3. By Lemma 3.1 with $\tau = h^{-1}$ we have

$$\begin{aligned}\|\boldsymbol{e}_q\|^2_{L^p(\Omega)} + h^{-1}\|e_u - e_{\widehat{u}}\|^2_{\partial\Omega_h} \le C(&|\langle e_u - e_{\widehat{u}}\,,\, (\boldsymbol{\sigma} - \Pi_V \boldsymbol{\sigma}) \cdot \boldsymbol{n}\rangle_{\partial\mathcal{T}_h}| \\ &+ |(|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\,,\, \boldsymbol{e}_q)_{\mathcal{T}_h}| \\ &+ |\langle h^{-1}(\Pi_W u - \Pi_M u)\,,\, e_u - e_{\widehat{u}}\rangle_{\partial\mathcal{T}_h}|) \\ := C(&T_1 + T_2 + T_3).\end{aligned}$$

We can control the three terms on the right hand side as:

$$\begin{aligned}T_1 &\le Ch^{-\frac{1}{2}}\|e_u - e_{\widehat{u}}\|_{\partial\mathcal{T}_h} h^{\frac{1}{2}} \|\boldsymbol{\sigma} - \Pi_V \boldsymbol{\sigma}\|_{\partial\mathcal{T}_h} \\ &\le Ch^{-\frac{1}{2}}\|e_u - e_{\widehat{u}}\|_{\partial\mathcal{T}_h} h \|\boldsymbol{\sigma}\|_{H^1(\Omega)}, \\ T_2 &\le \|\boldsymbol{e}_q\|_{L^p(\Omega)} \||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\|_{L^q(\Omega)} \\ &\le C\|\boldsymbol{e}_q\|_{L^p(\Omega)} \|\boldsymbol{q} - \Pi_V \boldsymbol{q}\|^{p-1}_{L^p(\Omega)} \\ &\le Ch^{t(p-1)}\|\boldsymbol{e}_q\|_{L^p(\Omega)} \|\boldsymbol{q}\|^{p-1}_{W^{t,p}(\Omega)}, \\ T_3 &\le Ch^{-\frac{1}{2}}\|e_u - e_{\widehat{u}}\|_{\partial\mathcal{T}_h} h^{-\frac{1}{2}} \|\Pi_W u - \Pi_M u\|_{\partial\mathcal{T}_h} \\ &\le Ch^{-\frac{1}{2}}\|e_u - e_{\widehat{u}}\|_{\partial\mathcal{T}_h} h^{-\frac{1}{2}} (\|u - \Pi_W u\|_{\partial\mathcal{T}_h} + \|u - \Pi_M u\|_{\partial\mathcal{T}_h}) \\ &\le Ch^{-\frac{1}{2}}\|e_u - e_{\widehat{u}}\|_{\partial\mathcal{T}_h} h \|u\|_{H^2(\Omega)}.\end{aligned}$$

Here for T_2 we used (2.8a) with $\sigma = 2 - p$ for the term $|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}$. Finally, with the Young's inequality together with above two inequalities we arrive at:

$$\|\boldsymbol{e}_q\|_{L^p(\Omega)} \le Ch(\|\boldsymbol{\sigma}\|_{H^1(\Omega)} + \|u\|_{H^2(\Omega)}) + Ch^{t(p-1)}\|\boldsymbol{q}\|^{p-1}_{W^{t,p}(\Omega)}.$$

Similarly, in the case of $p > 2$, taking $\sigma = p - 2$, $\tau = h^{-1}$ in Lemma 3.1 and together with (2.9b) we have

$$\begin{aligned}\|\boldsymbol{e}_q\|^p_{L^p(\Omega)} + h^{-1}\|e_u - e_{\widehat{u}}\|^2_{\partial\Omega_h} &\le |\boldsymbol{e}_q|_{(\Pi_V \boldsymbol{q}, p, p)} + h^{-1}\|e_u - e_{\widehat{u}}\|^2_{\partial\Omega_h} \\ &\le C(T_1 + T_2 + T_3). \end{aligned} \tag{3.3}$$

For T_1, T_3 we use the same estimates in the first case. For T_2, by Hölder's inequality and (2.8a) with $\sigma = 0$ we have

$$\begin{aligned}T_2 &\le \|\boldsymbol{e}_q\|_{L^p(\Omega)} \||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\|_{L^q(\Omega)} \\ &\le \|\boldsymbol{e}_q\|_{L^p(\Omega)} \||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\Pi_V \boldsymbol{q}|^{p-2}\Pi_V \boldsymbol{q}\|_{L^q(\Omega)} \\ &\le \|\boldsymbol{e}_q\|_{L^p(\Omega)} \||\boldsymbol{q} - \Pi_V \boldsymbol{q}|(|\boldsymbol{q}| + |\Pi_V \boldsymbol{q}|)^{p-2}\|_{L^q(\Omega)}.\end{aligned}$$

For the last term in the above inequality, we can further apply the Hölder's inequality as:

$$\begin{aligned}
\|\|\boldsymbol{q}-\Pi_V\boldsymbol{q}|(|\boldsymbol{q}|+|\Pi_V\boldsymbol{q}|)^{p-2}\|_{L^q(\Omega)} &= \Big(\int_\Omega |\boldsymbol{q}-\Pi_V\boldsymbol{q}|^q(|\boldsymbol{q}|+|\Pi_V\boldsymbol{q}|)^{q(p-2)}d\boldsymbol{x}\Big)^{\frac{1}{q}} \\
&= \Big(\int_\Omega |\boldsymbol{q}-\Pi_V\boldsymbol{q}|^q(|\boldsymbol{q}|+|\Pi_V\boldsymbol{q}|)^{p-q}d\boldsymbol{x}\Big)^{\frac{1}{q}} \\
&\le \Big(\big(\int_\Omega |\boldsymbol{q}-\Pi_V\boldsymbol{q}|^{q\cdot\frac{p}{q}}d\boldsymbol{x}\big)^{\frac{q}{p}}\big(\int_\Omega (|\boldsymbol{q}|+|\Pi_V\boldsymbol{q}|)^{(p-q)\cdot\frac{p}{p-q}}d\boldsymbol{x}\big)^{\frac{p-q}{p}}\Big)^{\frac{1}{q}} \\
&= \|\boldsymbol{q}-\Pi_V\boldsymbol{q}\|_{L^p(\Omega)}\||\boldsymbol{q}|+|\Pi_V\boldsymbol{q}|\|_{L^p(\Omega)}^{p-2} \\
&\le Ch^t\|\boldsymbol{q}\|_{W^{t,p}(\Omega)}. \qquad (3.4)
\end{aligned}$$

With the above estimates for T_1, T_2, T_3 and (3.3), we have the error estimates for the case $p > 2$ as:

$$\|\boldsymbol{e}_q\|_{L^p(\Omega)} \le Ch^{\frac{2}{p}}(\|\boldsymbol{\sigma}\|_{H^1(\Omega)} + \|u\|_{H^2(\Omega)}) + Ch^{\frac{t}{p-1}}\|\boldsymbol{q}\|_{W^{t,p}(\Omega)}. \qquad (3.5)$$

This completes the proof for (3.2).

□

4 Error Estimates for Simplicial Mesh

In this section, we present an alternative error estimates for the scheme (2.1) with the assumption that $\mathcal{T}_h$ is a conforming triangulation consisting of triangle elements. Under this setting, we are able to compare the HDG solution $(u_h, \widehat{u}_h)$ with the conforming solution u_c in [2]. With a careful choice on the stabilization parameter τ we can show that the HDG solution has the same convergence property as the CG solution u_c. Let us first recall the definition and the convergence properties of the CG solution u_c. We define

$$W_h^c(g) := \{w \in C^0(\Omega) \cap W_h, w|_{\partial\Omega} = g_h\},$$

where g_h is chosen to be conforming interpolant of g on $\partial\Omega$, see [2]. For the sake of simplicity, in the following analysis, we simply assume $g_h = g$ so that the error due to the interpolant is neglected. Define the CG solution $u_c \in W_h^c(g)$ which satisfies:

$$(|\nabla u_c|^{p-2}\nabla u_c, \nabla w) = (f, w) \quad \text{for all} \quad v \in W_h^c(0). \qquad (4.1)$$

In [2] the authors showed that the above approximation has the following convergence results:

Theorem 4.1 *If $u \in W^{2,p}(\Omega)$, then we have*

$$\|u - u_c\|_{W^{1,p}(\Omega)} \le \begin{cases} Ch^{\frac{p}{2}} & 1 < p \le 2 \\ Ch^{\frac{2}{p}} & p > 2 \end{cases} \qquad (4.2)$$

In addition, when $p \in (1, 2]$ with stronger regularity assumption $u \in W^{3,1}(\Omega) \cap C^{2,\frac{2-p}{p}}(\overline{\Omega})$, then we have

$$\|u - u_c\|_{W^{1,p}(\Omega)} \le Ch.$$

It is easy to see that if we set $\widehat{u}_c = u_c|_{\mathcal{E}_h}$ on $\mathcal{E}_h$, we have

$$\boldsymbol{Q}_h(u_c, \widehat{u}_c) = -\nabla u_c := \boldsymbol{q}_c.$$

We define

$$e_u^c = u_c - u_h, \quad e_{\widehat{u}}^c = \widehat{u}_c - \widehat{u}_h, \quad \boldsymbol{e}_q^c = \boldsymbol{q}_c - \boldsymbol{q}_h.$$

In addition, for the vector unknown $\boldsymbol{\sigma}$ we equip it with the lowest order **BDM** projection $\Pi_{BDM}\boldsymbol{\sigma}$. Next we show that with sufficiently large stabilization parameter τ, the HDG solution can have the same convergence property as the CG solution u_c. We first establish the following energy estimate which is similar to Lemma 3.1.

Lemma 4.2 *If $\mathcal{T}_h$ is a conforming simplicial triangulation, we have*

$$\begin{aligned} &C(|\boldsymbol{e}_q^c|_{(\boldsymbol{q}_c,p,2+\delta)} + \tau\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}^2) \\ &\quad \le -\langle e_u^c - e_{\widehat{u}}^c,\ (\boldsymbol{\sigma} - \Pi_{BDM}\boldsymbol{\sigma})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\,,\ \boldsymbol{e}_q^c)_{\mathcal{T}_h}. \end{aligned} \tag{4.3}$$

Proof Notice that $(u_c, \widehat{u}_c) \in W_h \times M_h$. Hence we can apply the same argument as in the proof of Lemma 3.1 with $(u_c, \widehat{u}_c)$ instead of $(\Pi_W u, \Pi_M u)$, $\Pi_{BDM}\boldsymbol{\sigma}$ instead of $\Pi_V\boldsymbol{\sigma}$ respectively. Consequently, we have the energy estimates as:

$$\begin{aligned} &C(|\boldsymbol{e}_q^c|_{(\boldsymbol{q}_c,p,2+\delta)} + \tau\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}^2) \\ &\quad \le -\langle e_u^c - e_{\widehat{u}}^c,\ (\boldsymbol{\sigma} - \Pi_{BDM}\boldsymbol{\sigma})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\,,\ \boldsymbol{e}_q^c)_{\mathcal{T}_h} \\ &\qquad + \langle\tau(u_c - \widehat{u}_c)\,,\ e_u^c - e_{\widehat{u}}^c\rangle_{\partial\mathcal{T}_h}. \end{aligned}$$

The proof is complete with the fact that $u_c - \widehat{u}_c = 0$ on $\partial\mathcal{T}_h$. □

Next we introduce the Oswald interpolation $\mathcal{I}_h : W_h \to W_h^c(g)$. Given a function $v_h \in W_h$, $\mathcal{I}_h v_h$ is prescribed at the Lagrangian nodes (interior vertices) on $\mathcal{E}_h \backslash \partial\Omega$ by the average of v_h at the nodes. For the nodes (vertices) on $\partial\Omega$, it takes the value of g_h. The interpolation has the following approximation result [18,30]:

Lemma 4.3 *For any $v_h \in W_h$ and any multi-index α with $|\alpha| = 0, 1$, the following estimate holds:*

$$\|D^\alpha(v_h - \mathcal{I}_h v_h)\|_{L^2(\mathcal{T}_h)}^2 \le C\left(\sum_{F\in\mathcal{E}_h^0} h_F^{1-2|\alpha|}\|[v_h]\|_{L^2(F)}^2 + \sum_{F\in\mathcal{E}_h^\partial} h_F^{1-2|\alpha|}\|g_h - v_h\|_{L^2(F)}^2\right).$$

For our final error estimates, we also need the following estimates:

Lemma 4.4 *We have:*

$$\|\boldsymbol{q}_h + \nabla\mathcal{I}_h u_h\|_{L^p(\mathcal{T}_h)} \le Ch^{-\frac{1}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h} \quad \text{for } p \in (1, 2], \tag{4.4a}$$

$$\|\boldsymbol{q}_h + \nabla\mathcal{I}_h u_h\|_{L^p(\mathcal{T}_h)} \le Ch^{\frac{2}{p}-\frac{3}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}, \quad \text{for } p \in (2, \infty). \tag{4.4b}$$

Proof We first prove the case $p = 2$. Applying integrating by parts on (2.1a) with $\boldsymbol{v}_1 = \boldsymbol{q}_h + \nabla u_h$, we arrive at:

$$\|\boldsymbol{q}_h + \nabla u_h\|_{L^2(\mathcal{T}_h)}^2 = \langle u_h - \widehat{u}_h\,,\ (\boldsymbol{q}_h + \nabla u_h)\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h}. \tag{4.5}$$

By the Cauchy–Schwartz inequality and the discrete trace inequality, we can obtain:

$$\|\boldsymbol{q}_h + \nabla u_h\|_{L^2(\mathcal{T}_h)} \le Ch^{-\frac{1}{2}}\|u_h - \widehat{u}_h\|_{\partial\mathcal{T}_h} = Ch^{-\frac{1}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}.$$

On the other hand, by Lemma 4.3, we know

$$\|\nabla u_h - \nabla \mathcal{I}_h u_h\|^2_{L^2(\mathcal{T}_h)} \le C \left(\sum_{F\in\mathcal{E}_h^0} h_F^{-1} \|[u_h]\|^2_{L^2(F)} + \sum_{F\in\mathcal{E}_h^\partial} h_F^{-1} \|g_h - u_h\|^2_{L^2(F)} \right)$$
$$\le C h^{-1} \|u_h - \widehat{u}_h\|^2_{\partial\mathcal{T}_h} = C h^{-1} \|e_u^c - e_{\widehat{u}}^c\|^2_{\partial\mathcal{T}_h}.$$

The last step is due to the fact that $\widehat{u}_h$ is single-valued on $\mathcal{E}_h$ and a triangle inequality. Finally, by the above two estimates and a simple triangle inequality we have

$$\|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|_{L^2(\mathcal{T}_h)} \le \|\boldsymbol{q}_h + \nabla u_h\|_{L^2(\mathcal{T}_h)} + \|\nabla u_h - \nabla \mathcal{I}_h u_h\|_{L^2(\mathcal{T}_h)} \le C h^{-\frac{1}{2}} \|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}.$$

For $p \in (1, 2]$, we simply apply the Sobolev inequality to have:

$$\|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|_{L^p(\mathcal{T}_h)} \le C \|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|_{L^2(\mathcal{T}_h)} \le C h^{-\frac{1}{2}} \|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}.$$

For $p \in (2, \infty)$, first we recall the discrete inverse inequality: $\|v\|_{L^p(K)} \le C h_K^{\frac{2}{p}-1} \|v\|_{L^2(K)}$. We have

$$\|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|^p_{L^p(\mathcal{T}_h)} = \sum_{K\in\mathcal{T}_h} \|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|^p_{L^p(K)} \le C \sum_{K\in\mathcal{T}_h} h_K^{p(\frac{2}{p}-1)} \|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|^p_{L^2(K)}$$
$$\le C h^{2-p} \sum_{K\in\mathcal{T}_h} \left(\|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|^2_{L^2(K)} \right)^{\frac{p}{2}}$$
$$\le C h^{2-p} \left(\sum_{K\in\mathcal{T}_h} \|\boldsymbol{q}_h + \nabla \mathcal{I}_h u_h\|^2_{L^2(K)} \right)^{\frac{p}{2}}.$$

The last step we used the fact that $x^{\frac{p}{2}} + y^{\frac{p}{2}} \le (x + y)^{\frac{p}{2}}$ when $p \ge 2$ with $x, y \ge 0$. The proof is completed with (4.5). □

We are ready to present our final error estimates.

Theorem 4.5 *If $\mathcal{T}_h$ is a conforming simplicial triangulation, with the same regularity assumption as in Theorem 4.1. For $p \in (1, 2]$ with $\tau = \mathcal{O}(h^{-3})$ we have*

$$\|\boldsymbol{q} - \boldsymbol{q}_h\|_{L^p(\Omega)} \le C(\|u - u_c\|_{W^{1,p}(\Omega)} + \mathcal{O}(h)).$$

For $p \in (2, \infty)$ with $\tau = \mathcal{O}(h^{\frac{8}{p}-5})$, we have

$$\|\boldsymbol{q} - \boldsymbol{q}_h\|_{L^p(\Omega)} \le C h^{\frac{2}{p}}.$$

Proof Notice that since the exact solution also satisfies the Eq. (4.1). This means that

$$(|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\,,\ \nabla w)_{\mathcal{T}_h} = 0, \quad \text{for all} \quad w \in W_h(0).$$

Taking $w = \mathcal{I}_h u_h - u_c \in W_h(0)$, we have

$$(|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\,,\ \nabla(\mathcal{I}_h u_h - u_c))_{\mathcal{T}_h} = 0.$$

Inserting this identity in the last term of (4.3) in Lemma 4.2 we have

$$\begin{aligned} &C(|\boldsymbol{e}_q^c|_{(\boldsymbol{q}_c,p,2+\delta)} + \tau\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}^2) \\ &\quad \le -\langle e_u^c - e_{\widehat{u}}^c,\, (\boldsymbol{\sigma} - \Pi_{BDM}\boldsymbol{\sigma})\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} - (|\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c,\, \boldsymbol{q}_h + \nabla_{\mathcal{I}_h} u_h)_{\mathcal{T}_h} \\ &\quad := S_1 + S_2. \end{aligned} \tag{4.6}$$

In the case $p \in (1, 2]$, by Lemma 4.2 with $\sigma = 2$ in (2.9a) with $(\boldsymbol{\psi}, \boldsymbol{\eta}) = (\Pi_V \boldsymbol{q}, \boldsymbol{e}_q)$ we have

$$\|\boldsymbol{e}_q\|_{L^p(\Omega)}^2 + \tau\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}^2 \le C(|\boldsymbol{e}_q^c|_{(\boldsymbol{q}_c,p,2+\delta)} + \tau\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}^2) \le C(S_1 + S_2). \tag{4.7}$$

Applying the Cauchy–Schwartz inequality for S_1 we have

$$\begin{aligned} S_1 &\le \tau^{\frac{1}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}\tau^{-\frac{1}{2}}\|(\boldsymbol{\sigma} - \Pi_{BDM}\boldsymbol{\sigma})\cdot\boldsymbol{n}\|_{\partial\mathcal{T}_h} \\ &\le C\tau^{\frac{1}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h} h^{\frac{1}{2}}\tau^{-\frac{1}{2}}\|\boldsymbol{\sigma}\|_{H^1(\Omega)}, \end{aligned} \tag{4.8}$$

For S_2, we first apply the Hölder's inequality to have

$$\begin{aligned} S_2 &\le \||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\|_{L^q(\Omega)}\|\boldsymbol{q}_h + \nabla_{\mathcal{I}_h} u_u\|_{L^p(\Omega)} \\ &\le C\||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\|_{L^q(\Omega)} h^{-\frac{1}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}. \end{aligned}$$

The later inequality is due to (4.4a). Further, by (2.8a) with $\sigma = 2 - p$ we have

$$\||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\|_{L^q(\Omega)} \le C\|\boldsymbol{q} - \boldsymbol{q}_c\|_{L^p(\Omega)}^{p-1}.$$

This implies that

$$S_2 \le Ch^{-\frac{1}{2}}\tau^{-\frac{1}{2}}\|\boldsymbol{q} - \boldsymbol{q}_c\|_{L^p(\Omega)}^{p-1}\tau^{\frac{1}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}. \tag{4.9}$$

Now if we combine the estimates (4.7), (4.8), (4.9) and with Young's inequality, we can obtain:

$$\|\boldsymbol{e}_q^c\|_{L^p(\Omega)}^2 \le C\left(h\tau^{-1}\|\boldsymbol{\sigma}\|_{H^1(\Omega)}^2 + h^{-1}\tau^{-1}\|\boldsymbol{q} - \boldsymbol{q}_c\|_{L^p(\Omega)}^{2(p-1)}\right).$$

If we take $\tau = \mathcal{O}(h^{-3})$, the above estimate gives

$$\|\boldsymbol{e}_q^c\|_{L^p(\Omega)} \le Ch.$$

Consequently, by Theorem 4.1 and triangle inequality, we have

$$\|\boldsymbol{q} - \boldsymbol{q}_h\|_{L^p(\Omega)} \le C(\|u - u_c\|_{W^{1,p}(\Omega)} + \mathcal{O}(h)).$$

Next we derive the estimates for $p > 2$. For the left hand side of (4.6), we take $\sigma = p - 2$ and apply (2.9a) to have:

$$\|\boldsymbol{e}_q^c\|_{L^p(\Omega)}^p + \tau\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}^2 \le S_1 + S_2. \tag{4.10}$$

For S_1 we use the same estimates as (4.8). For S_2, by the Hölder's inequality and (4.4b) we have

$$\begin{aligned} S_2 &\le \||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\|_{L^q(\Omega)}\|\boldsymbol{q}_h + \nabla_{\mathcal{I}_h} u_u\|_{L^p(\Omega)} \\ &\le C\||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\|_{L^q(\Omega)} h^{\frac{2}{p}-\frac{3}{2}}\|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}. \end{aligned} \tag{4.11}$$

With the same argument as (3.4), we can obtain:

$$\||\boldsymbol{q}|^{p-2}\boldsymbol{q} - |\boldsymbol{q}_c|^{p-2}\boldsymbol{q}_c\|_{L^q(\Omega)} \le C\|\boldsymbol{q} - \boldsymbol{q}_c\|_{L^p(\Omega)}\||\boldsymbol{q}| + |\boldsymbol{q}_c|\|_{L^p(\Omega)}^{p-2} \le C\|\boldsymbol{q} - \boldsymbol{q}_c\|_{L^p(\Omega)}.$$

Inserting this inequality into (4.11) we have

$$S_2 \leq C\tau^{-\frac{1}{2}} h^{\frac{2}{p}-\frac{3}{2}} \|\boldsymbol{q}-\boldsymbol{q}_c\|_{L^p(\Omega)} \tau^{\frac{1}{2}} \|e_u^c - e_{\widehat{u}}^c\|_{\partial\mathcal{T}_h}. \tag{4.12}$$

Combining the estimates (4.10), (4.8), (4.12) and Young's inequality we can obtain:

$$\begin{aligned}\|\boldsymbol{e}_q^c\|_{L^p(\Omega)}^p &\leq C\left(h\tau^{-1}\|\boldsymbol{\sigma}\|_{H^1(\Omega)}^2 + h^{\frac{4}{p}-3}\tau^{-1}\|\boldsymbol{q}-\boldsymbol{q}_c\|_{L^p(\Omega)}^2\right)\\ &\leq C\left(h\tau^{-1}\|\boldsymbol{\sigma}\|_{H^1(\Omega)}^2 + \mathcal{O}(h^{\frac{8}{p}-3}\tau^{-1})\right).\end{aligned}$$

The last step is due to Theorem 4.1. Finally, if we take $\tau = \mathcal{O}(h^{\frac{8}{p}-5})$ the above estimate becomes

$$\|\boldsymbol{e}_q^c\|_{L^p(\Omega)}^p \leq C\left(\mathcal{O}(h^{6-\frac{8}{p}}) + \mathcal{O}(h^2)\right) \sim \mathcal{O}(h^2).$$

Consequently, by the above estimate and Theorem 4.1 we have

$$\|\boldsymbol{q}-\boldsymbol{q}_h\|_{L^p(\Omega)} \leq \|\boldsymbol{q}-\boldsymbol{q}_c\|_{L^p(\Omega)} + \|\boldsymbol{e}_q^c\|_{L^p(\Omega)} \leq Ch^{\frac{2}{p}}.$$

This completes the proof. □

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